Churen (Chu) Shao

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Work History

Affirm - Staff Software Engineer
(Mar 2021 - Current)

- Tech lead across the Pricing and Incentives Engineering Team.
- Drive roadmap planning in partnership with product, analytics, and marketing stakeholders.
- Define long-term technical architecture and cross-product engineering goals for pricing and incentives.
- Architected AdaptAI, Affirm's AI-powered promotions platform, coordinating across marketplace, upfunnel, credit, and checkout teams. Now live across nearly all Affirm surfaces with ˜10% YoY improvement in end-to-end checkout conversion.
- Built a reinforcement learning based multi-armed bandit experimentation system for dynamic pricing configurations, serving as the foundation for BoostAI and enabling automatic A/B testing of financing offers across merchants.
- Designed the core pricing configuration bundle system, delivering over 100x performance improvement.
- Owned and completed a 4-year migration decoupling pricing from a monolith, requiring full interface rewrites across pricing and checkout systems.
- Defined and decoupled key pricing domain RPC endpoints from other systems and established internal SLAs.
- Mentor engineers across levels, partnering with managers to structure projects that target individual growth areas.
- Identify and close engineering velocity gaps across the team, from tooling and automation to process improvements.

Udemy - Senior Staff Software Engineer
(Jan 2015–Feb 2021)

- Tech lead on the pricing team supporting growth marketing, pricing, and discounts across the Udemy marketplace.
- Architected a multi-strategy pricing engine with ML-based optimization, generating 8-10% revenue lift.
- Designed and launched a new subscription platform, coordinating across engineering and product stakeholders.
- Built a targeting system to personalize pricing and promotions for any user attribute, backed by a bespoke Redis store.
- Designed and implemented a fully integrated and scalable affiliate network framework for on-boarding affiliates.
- Scaled In App Purchases on iOS and Android from 500 to over 150,000 courses, extending checkout to mobile platforms.
- Launched purchasing power parity pricing, enabling localized price points across 11 countries.
- Mentored interns and led design discussions in structuring a dynamic Google Adwords bid modifier platform.
- Redesigned the instructor coupon creation flow in React, integrating asynchronously with the pricing engine.
- Unified currency formatting across all systems for international currencies and locales, standardizing checkout display.
- Led stakeholder discussions translating pricing and promotions product requirements into technical design documents.
- Shipped internal tools supporting consumer marketing and promotional campaign operations.
- Migrated legacy PHP and Javascript pricing and checkout code to Django and Angular with over 95% test coverage.

CBS Interactive - Software Engineer
(Jun 2013–Jan 2015)

- Replatformed B2B websites with 23 million monthly views, including ZDNet, Smartplanet, and TechRepublic.
- Acted as west coast point of contact for managing deploys, fixing blockers, and running data migrations.
- Coded in various industry standard languages, including Spring Boot Java, Symfony2 PHP, Twig, HTML5, and Javascript.
- Extended and optimized APIs through Solr endpoints instead of MySQL DB calls, improving search indexer performance and scalability.
- Worked with a global team, answering questions and managing projects with Singapore and UK.

BlueHornet/Digital River - Software Engineer Intern
(Jun 2012–Sept 2012)

- Summer internship focused on adding a custom number field for utilization in the Blue Hornet email application.
- Programmed in PHP and worked with Zend framework, front-end UI, and Oracle 11g Database.
- Implemented a feature that is used to send personalized data to millions of subscribers.

Ocean Observatories Initiative - Systems and Network Support
(Jun 2011–Jun 2013)

- Managed a network of over 50 different virtual servers through monitoring tools and Python/Ruby/shell scripting.
- Worked with various levels of systems operations including hardware diagnosis and hypervisor management.

Publications

Enrico Tanuwidjaja, Derek Huynh, Kirsten Koa, Calvin Nguyen, **Churen Shao**, Patrick Torbett, Colleen Emmenegger, and Nadir Weibel. Chroma: A Wearable Augmented-Reality Solution for Color-Blindness. In *Proceedings of **Ubicomp 2014**, ACM International Joint Conference on Pervasive and Ubiquitous Computing*, ACM, New York, NY, USA, 799-810.
Nominated for Best Paper in Assistive Devices at Ubicomp 2014

Education

UC San Diego - Computer Science B.S. • June 2014
Georgia Institute of Technology - Computer Science M.S. • December 2019